

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

An End-to-End Data Analytics Project

Python (Pandas) · PostgreSQL (SQL) · Power BI

INTERNSHIP PROJECT REPORT

Submitted as part of Data Analyst Internship

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1. Introduction

Retail businesses collect large volumes of data about their customers every day - what they buy, how much they spend, how often they return, and how they respond to discounts. Making sense of this data is essential for improving marketing decisions, customer retention, and revenue growth.

This project analyzes a retail customer shopping dataset to understand purchasing patterns, customer segments, and revenue drivers. It was completed as a self-driven internship project to practice the full data analytics workflow: cleaning raw data, engineering useful features, answering business questions with SQL, and building an interactive dashboard for visual reporting.

1.1 Objective

The objective of this project is to analyze customer shopping behavior in order to answer key business questions such as:

- Which customer segments generate the most revenue?
- Do discounts and subscriptions actually influence spending?
- Which products and categories perform best?
- How can customers be segmented by loyalty (New, Returning, Loyal)?
- What insights can support better marketing and inventory decisions?

1.2 Tools & Technologies Used

Tool	Purpose
Python (Pandas, Jupyter Notebook)	Data cleaning, preprocessing, feature engineering
PostgreSQL	Storing cleaned data and running SQL business queries
SQL	Answering business questions using aggregation, window functions, CTEs
SQLAlchemy / psycopg2	Connecting Python to PostgreSQL and loading data
Power BI	Building an interactive dashboard for visual insights

2. Dataset Description

The dataset used is a retail customer shopping behavior dataset containing 3,900 records and 18 columns, covering customer demographics, purchase details, and shopping preferences.

Column	Description
Customer ID	Unique identifier for each customer
Age, Gender	Customer demographics
Item Purchased, Category	Product bought and its category
Purchase Amount (USD)	Value of the transaction
Location, Size, Color, Season	Purchase context and product attributes
Review Rating	Customer's rating of the product (1-5)
Subscription Status	Whether the customer is a subscriber
Shipping Type	Standard, Express, Free Shipping, Next Day Air, etc.
Discount Applied, Promo Code Used	Whether a discount/promo was used
Previous Purchases	Number of past purchases by the customer
Payment Method, Frequency of Purchases	How and how often the customer buys

Key dataset facts:

- Total records: 3,900 | Total columns: 18
- Categories: Clothing (1,737), Accessories (1,240), Footwear (599), Outerwear (324)
- Gender split: Male (2,652), Female (1,248)
- Age range: 18 to 70 years
- Purchase amount range: \$20 to \$100 per transaction
- Missing values: 37 missing entries in 'Review Rating' (the only column with nulls)

3. Project Workflow

The project followed a standard end-to-end analytics pipeline:

1. Data Loading - Imported the raw CSV file into a Pandas DataFrame in Jupyter Notebook.
2. Data Exploration - Used `.info()`, `.describe()`, and `.isnull().sum()` to understand structure, data types, and missing values.
3. Data Cleaning - Handled missing review ratings and standardized column names.
4. Feature Engineering - Created new columns (`age_group`, `purchase_frequency_days`) and removed a redundant column.
5. Database Loading - Pushed the cleaned dataset into a PostgreSQL database using SQLAlchemy.

6. SQL Analysis - Wrote 10 business-focused SQL queries to extract insights.
7. Dashboard Building - Visualized the findings in an interactive Power BI dashboard.

4. Data Cleaning & Preprocessing

4.1 Handling Missing Values

The 'Review Rating' column had 37 missing values. Instead of dropping these rows (which would lose other valid data), missing ratings were filled using the median review rating within each product category. This preserves category-level rating patterns better than a single global average.

4.2 Standardizing Column Names

Column names were converted to lowercase and spaces were replaced with underscores (e.g., 'Purchase Amount (USD)' became 'purchase_amount') to make the dataset easier to query in SQL and consistent to work with in Python.

4.3 Feature Engineering

Two new columns were engineered to support deeper analysis:

- `age_group`: Customers were split into four equal-sized groups (Young Adult, Adult, Middle-aged, Senior) using quantile-based binning (`pd.qcut`) on the `Age` column.
- `purchase_frequency_days`: The text-based 'Frequency of Purchases' field (e.g., Weekly, Monthly, Quarterly) was mapped to an approximate number of days between purchases, making it usable in numeric calculations.

During exploration, it was also found that the 'Discount Applied' and 'Promo Code Used' columns were identical for every record. Since one column was fully redundant, 'Promo Code Used' was dropped to simplify the dataset.

4.4 Loading into PostgreSQL

The cleaned DataFrame was loaded into a PostgreSQL database (`customer_behavior`) as a table named 'customer' using SQLAlchemy's `to_sql()` function, making the data available for SQL-based analysis.

```

1 from sqlalchemy import create_engine
2
3 username = "postgres"
4 password = "*****"
5 host = "localhost"
6 port = "5432"
7 database = "customer_behavior"
8
9 engine = create_engine(
10     f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
11 )
12
13 table_name = 'customer'
14 df.to_sql(table_name, engine, if_exists='replace', index=False)
15
16 print(f"Data successfully loaded into table '{table_name}'.")

```

Figure 1: Connecting to PostgreSQL and loading the cleaned DataFrame (Jupyter Notebook)

4.5 Code Snapshots

The screenshots below are taken directly from the project's Jupyter Notebook, showing the cleaning and feature engineering steps described above.

```

1 import pandas as pd
2
3 df = pd.read_csv(r"customer_shopping_behavior.csv")
4
5 # Explore structure and missing values
6 df.info()
7 df.describe(include='all')
8 df.isnull().sum()
9
10 # Fill missing review ratings using category-level median
11 df['Review Rating'] = df.groupby('Category')['Review Rating']\
12     .transform(lambda x: x.fillna(x.median()))
13
14 # Standardize column names
15 df.columns = df.columns.str.lower().str.replace(' ', '_')
16 df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

```

Figure 2: Handling missing values and standardizing column names

```

1 # Feature engineering: bucket customers into age groups
2 labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
3 df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
4
5 # Convert purchase frequency text into numeric days
6 frequency_mapping = {
7     'Fortnightly': 14, 'Weekly': 7, 'Monthly': 30,
8     'Quarterly': 90, 'Bi-Weekly': 14, 'Yearly': 365,
9     'Every 3 Months': 90
10 }
11 df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
12
13 # Drop redundant column (identical to discount_applied)
14 df = df.drop('promo_code_used', axis=1)

```

Figure 3: Feature engineering - age_group, purchase_frequency_days, and dropping the redundant column

5. SQL Business Analysis & Key Insights

Ten business questions were answered using SQL queries involving aggregation, filtering, window functions (ROW_NUMBER), and common table expressions (CTEs). The queries and their real results from the dataset are summarized below.

```

1 -- Q8: Top 3 most purchased products within each category
2 WITH item_counts AS (
3     SELECT category,
4           item_purchased,
5           COUNT(customer_id) AS total_orders,
6           ROW_NUMBER() OVER (
7               PARTITION BY category
8               ORDER BY COUNT(customer_id) DESC
9           ) AS item_rank
10    FROM customer
11   GROUP BY category, item_purchased
12 )
13 SELECT item_rank, category, item_purchased, total_orders
14 FROM item_counts
15 WHERE item_rank <= 3;

```

Figure 4: Example SQL query (Q8) using a CTE and ROW_NUMBER() window function

Q1. Revenue by gender

Question: What is the total revenue generated by male vs. female customers?

Gender	Total Revenue (USD)
Male	157,890
Female	75,191

Insight: Male customers contributed roughly 68% of total revenue, more than double that of female customers, despite the average spend per transaction being similar - this is driven mainly by a larger male customer base (2,652 vs 1,248).

Q2. Discount users spending above average

Question: Which customers used a discount but still spent more than the average purchase amount?

Insight: 839 out of 3,900 transactions (about 21.5%) involved a discount and still exceeded the average purchase amount of \$59.76. This shows that discounts are not always used only for budget purchases - many discount-seeking customers are also high-value spenders.

Q3. Top 5 highest-rated products

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

Insight: Accessories and footwear items dominate the top-rated products, suggesting stronger customer satisfaction in these categories compared to core clothing items.

Q4. Standard vs. Express shipping spend

Shipping Type	Avg. Purchase Amount (USD)
Express	60.48
Standard	58.46

Insight: Customers using Express shipping spend slightly more on average (\$60.48) than those using Standard shipping (\$58.46), suggesting a small willingness among higher-spending customers to pay for faster delivery.

Q5. Subscribers vs. non-subscribers

Status	Customers	Avg. Spend	Total Revenue
Non-subscriber	2,847	\$59.87	\$170,436
Subscriber	1,053	\$59.49	\$62,645

Insight: Surprisingly, subscribers do not spend more per transaction than non-subscribers (\$59.49 vs \$59.87). Subscription status alone is not a strong predictor of higher individual spend, although non-subscribers, being a much larger group, drive the majority of total revenue.

Q6. Products with highest discount usage

Rank	Product	% Purchases with Discount
1	Hat	50.0%
2	Sneakers	49.7%
3	Coat	49.1%
4	Sweater	48.2%
5	Pants	47.4%

Insight: Around half of all Hat and Sneakers purchases involved a discount, indicating these items may be more price-sensitive or are used strategically as promotional products.

Q7. Customer segmentation by loyalty

Segment	Definition	Number of Customers
New	1 previous purchase	83
Returning	2-10 previous purchases	701
Loyal	More than 10 previous purchases	3,116

Insight: The vast majority of customers (about 80%) fall into the 'Loyal' segment with more than 10 previous purchases, indicating a strong repeat-customer base. Truly new customers make up only about 2% of the dataset.

Q8. Top 3 products per category

Category	Top 3 Products
Clothing	Blouse, Pants, Shirt
Accessories	Jewelry, Sunglasses, Belt
Footwear	Sandals, Shoes, Sneakers
Outerwear	Jacket, Coat

Insight: Using SQL window functions (ROW_NUMBER), the best-selling products within each category were identified, giving a clear view of which specific items drive volume in each product line - useful for inventory prioritization.

Q9. Repeat buyers and subscription likelihood

Subscription Status	Repeat Buyers (>5 previous purchases)
No	2,518
Yes	958

Insight: Repeat buyers are more likely to NOT be subscribers (2,518 vs 958), suggesting that customer loyalty (measured by repeat purchases) develops independently of formal subscription programs in this dataset.

Q10. Revenue contribution by age group

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Insight: Revenue is fairly evenly distributed across all age groups, with Young Adults contributing slightly more than others. This suggests the product range appeals broadly across ages rather than being concentrated in one demographic.

6. Power BI Dashboard

An interactive Power BI dashboard (Customer_Behavior_Dashboard.pbix) was built on top of the cleaned dataset to make the findings accessible to non-technical stakeholders. The dashboard translates the SQL insights above into visual form, including:

- Revenue breakdown by gender, category, and age group
- Customer segmentation view (New / Returning / Loyal)
- Discount usage and its relationship with spending
- Top-performing products and categories
- Filters/slicers for interactive exploration by season, location, and payment method

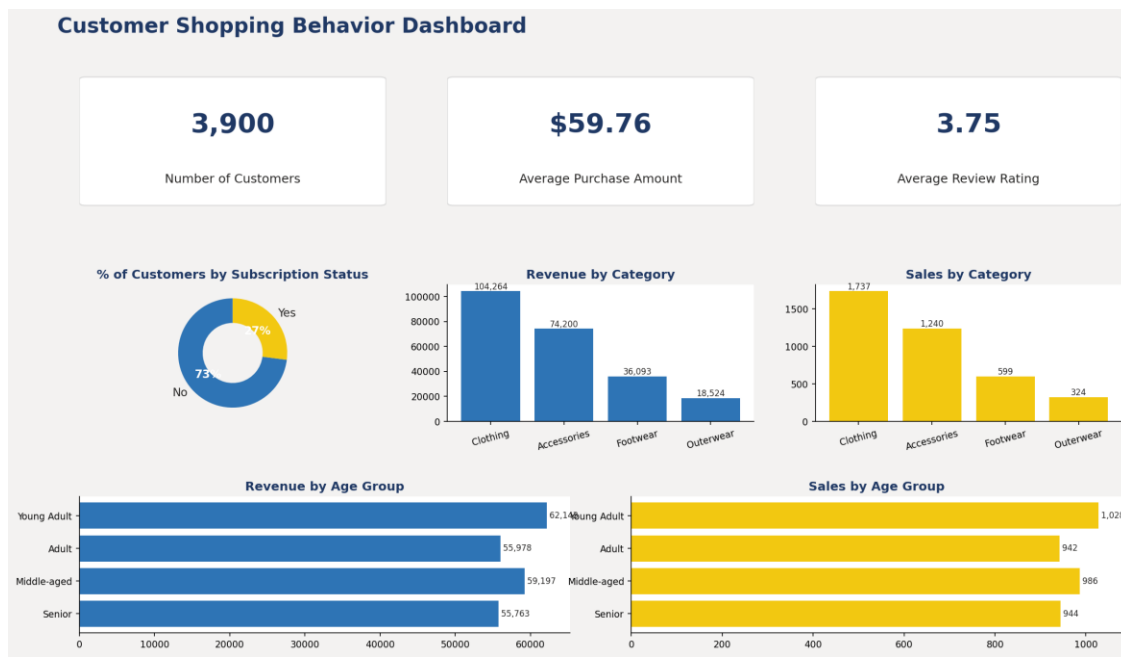


Figure 5: Dashboard recreation based on the actual Power BI report layout (Customer_Behavior_Dashboard.pbix) and dataset values - KPI cards, subscription split, revenue/sales by category, and revenue/sales by age group

Note: The visual above is a faithful recreation of the dashboard's actual layout, visuals, and field mappings (extracted directly from the .pbix file) rendered with the same underlying data, since opening the live .pbix file requires Power BI Desktop.

Dashboards like this allow business teams to explore the data themselves without needing to write SQL, making the analysis more actionable day to day.

7. Key Findings Summary

- Male customers generate significantly more total revenue, mainly due to a larger customer base, not higher per-transaction spend.
- About 80% of customers are 'Loyal' (10+ previous purchases), showing strong repeat business.
- Subscription status does not correlate with higher average spend or more repeat purchases - the subscription program may need to be re-evaluated or better targeted.
- Discounts are used broadly across both high and low spenders, with items like Hats and Sneakers being the most frequently discounted.
- Revenue is spread fairly evenly across age groups, indicating broad market appeal.

8. Challenges Faced

- Identifying and cleanly handling missing values in the Review Rating column without losing useful data from other columns.
- Recognizing that 'Discount Applied' and 'Promo Code Used' were fully redundant columns before deciding to drop one.
- Mapping inconsistent text categories (e.g., typos like 'Fortnightly') in the Frequency of Purchases column to numeric day values.
- Translating pandas-based exploration into equivalent, well-structured SQL queries using joins-free aggregation, CTEs, and window functions.
- Designing a Power BI dashboard that is both visually clear and genuinely useful for business decision-making.

9. Learnings & Skills Gained

- Practical experience with the full analytics pipeline: raw data to database to dashboard.
- Hands-on practice with Pandas for data cleaning and feature engineering.
- Writing SQL queries using GROUP BY, subqueries, CTEs, and window functions (ROW_NUMBER) to solve real business questions.
- Connecting Python to PostgreSQL using SQLAlchemy for an end-to-end data pipeline.
- Building an interactive Power BI dashboard to communicate insights to a non-technical audience.
- Developing a business mindset - translating raw data into insights that could genuinely inform marketing or inventory decisions.

10. Conclusion

This project provided practical, hands-on experience with the complete data analytics workflow used in real-world business settings: cleaning raw data, engineering meaningful features, storing data in a relational database, extracting insights using SQL, and communicating those insights visually through Power BI.

The analysis revealed several actionable insights - from the surprising lack of impact of subscriptions on spending, to the strong base of loyal repeat customers - that could help a retail business fine-tune its marketing, loyalty programs, and inventory strategy. As a fresher project, it strengthened both technical skills (Python, SQL, Power BI) and the ability to think from a business decision-making perspective.

Project Files: first_project.ipynb (data cleaning & feature engineering) · customer_shopping_behavior.sql (SQL business queries) · Customer_Behavior_Dashboard.pbix (Power BI dashboard) · customer_shopping_behavior.csv (raw dataset)